NEET UG PHYSICS SYLLABUS

Mechanical Properties of Fluids

Complete Conceptual Tutorial, Worked Examples, and NEET UG Practice Assessment

1. Learning Objectives & Chapter Overview

Liquids and gases can flow and are therefore collectively called **fluids**. This fundamental ability to flow distinguishes liquids and gases from solids. Unlike a solid, a fluid has no definite shape of its own. Solids and liquids have a fixed volume under atmospheric pressure, whereas a gas expands to fill the entire volume of its container.

Key Learning Objectives for NEET UG:

- Understand fluid pressure as a scalar quantity, Pascal's law, hydrostatic paradox, gauge pressure, and hydraulic machines.
- Analyze steady (streamline) flow and apply the Equation of Continuity ($Av = \text{constant}$) for incompressible fluids.
- Master Bernoulli's Principle, Torricelli's law of efflux, the Venturi-type constriction effect, Magnus effect, and dynamic lift on aerofoils.
- Quantify internal fluid friction using the coefficient of viscosity (η), Poiseuille flow concepts, Stokes' law ($F = 6\pi\eta a v$), and terminal velocity.
- Evaluate surface energy, surface tension (S), angle of contact (θ), excess pressure inside drops ($\Delta P = 2S/r$) and soap bubbles ($\Delta P = 4S/r$), and capillary rise or fall.

Why do fluids flow? Shear stress can change the shape of a solid while keeping its volume fixed. The key mechanical property of fluids is that they offer very little resistance to shear stress; their shape changes continuously by the application of even a very small shear stress. The shearing stress of fluids is about a million times smaller than that of solids. Furthermore, solids and liquids have much lower compressibility compared to gases, so liquids are largely treated as incompressible.

2. Pressure and Hydrostatics

2.1 Definition of Pressure and Density

When an object is submerged in a fluid at rest, the fluid exerts a force on its surface. This force is always **normal (perpendicular)** to the object's surface. If there were a component of force parallel to the surface, the object would also exert a parallel force on the fluid by Newton's third law, causing the fluid to flow parallel to the surface. Since the fluid is at rest, no parallel component can exist.

If F is the magnitude of the normal force acting on a surface of area A, the **average pressure** P_{av} and the **instantaneous pressure** P at a point are defined as:

$$P_{\text{av}} = \frac{F}{A} \qquad \text{and} \qquad P = \lim_{\Delta A \to 0} \frac{\Delta F}{\Delta A}$$

- **Nature:** Pressure is a **scalar quantity**. It is the component of the force normal to the area under consideration—not the vector force—that appears in the numerator.
- **Dimensions & SI Unit:** Dimensions are $[\mathrm{M\,L^{-1}\,T^{-2}}]$. The SI unit is $\mathrm{N\,m^{-2}}$, named **pascal** (Pa).
- **Common Pressure Units:**
  - $1\,\mathrm{atm} = 1.013 \times 10^5\,\mathrm{Pa}$ (pressure exerted by the atmosphere at sea level)
  - $1\,\mathrm{bar} = 10^5\,\mathrm{Pa}$
  - $1\,\mathrm{torr} = 1\,\mathrm{mm\ of\ Hg} = 133\,\mathrm{Pa} = 0.133\,\mathrm{kPa}$

For a fluid of mass m occupying volume V, its **density** ρ is defined as:

$$\rho = \frac{m}{V}$$

Density is a positive scalar quantity with dimensions $[\mathrm{M\,L^{-3}}]$ and SI unit $\mathrm{kg\,m^{-3}}$. The density of water at $4\,^{\circ}\mathrm{C}$ ($277\,\mathrm{K}$) is $1.0 \times 10^{3}\,\mathrm{kg\,m^{-3}}$. The **relative density** (or specific gravity) of a substance is the dimensionless ratio of its density to the density of water at $4\,^{\circ}\mathrm{C}$:

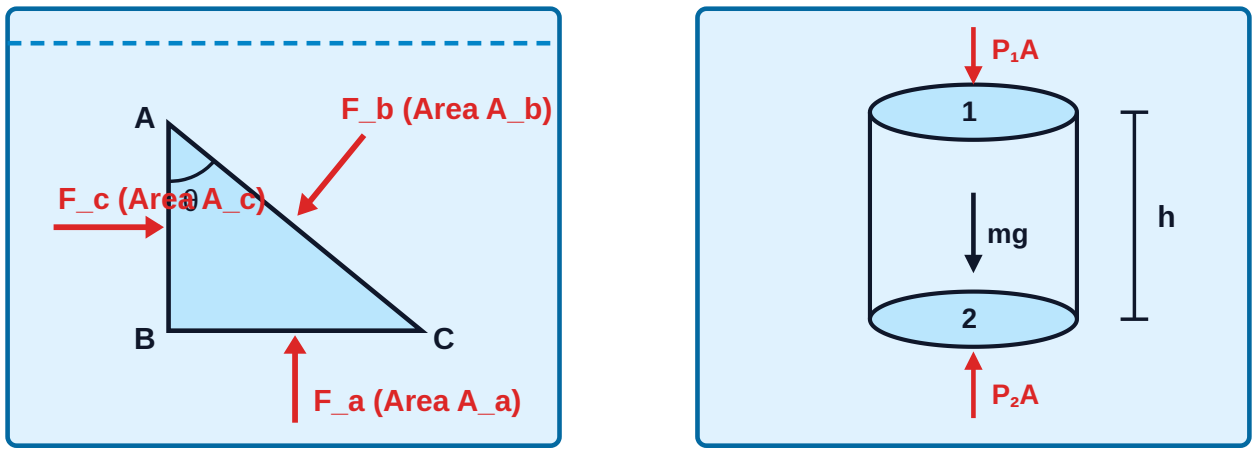
$$\text{Relative Density} = \frac{\rho_{\text{substance}}}{\rho_{\text{water at } 4\,^{\circ}\mathrm{C}}}$$

Densities of Common Fluids at Standard Temperature (0 °C) and Pressure (1 atm)

Fluid	Density ρ ($\mathrm{kg\,m^{-3}}$)
Water	1.00×10^{3}
Sea water	1.03×10^{3}
Mercury	13.6×10^{3}
Ethyl alcohol	0.806×10^{3}
Whole blood	1.06×10^{3}
Air	1.29
Oxygen	1.43
Hydrogen	9.0×10^{-2}
Interstellar space	$\approx 10^{-20}$

2.2 Pascal's Law

Blaise Pascal observed that the pressure in a fluid at rest is the same at all points if they are at the same height. Consider a very small right-angled prismatic fluid element ABC-DEF submerged inside a fluid at rest. Because the element is infinitesimally small, every part of it is at the same depth, and the effect of gravity can be ignored.



Tutorial Diagram 1: (Left) Equilibrium of a prismatic fluid element proving isotropic pressure. (Right) Vertical cylindrical fluid element of height h under gravity.

Let the normal forces exerted by the surrounding fluid on the faces of areas A_a (bottom), A_b (slanted face), and A_c (vertical face) be F_a, F_b, and F_c, respectively. By mechanical equilibrium and geometry:

$$\begin{aligned} F_b \sin\theta &= F_c \quad \text{and} \quad F_b \cos\theta = F_a \\ A_b \sin\theta &= A_c \quad \text{and} \quad A_b \cos\theta = A_a \end{aligned}$$

$$\implies \frac{F_b}{A_b} = \frac{F_c}{A_c} = \frac{F_a}{A_a} \implies P_b = P_c = P_a$$

Thus, pressure exerted at a point in a fluid at rest is the same in all directions. Furthermore, in the absence of flow, the pressure in a fluid must be the same everywhere in a horizontal plane.

2.3 Variation of Pressure with Depth & Hydrostatic Paradox

Consider a cylindrical element of fluid of base area A and height h between point 1 (top, pressure P_1) and point 2 (bottom, pressure P_2). Since the fluid is at rest, the upward force $P_2 A$ balances the downward force $P_1 A$ plus the weight $mg = (\rho A h)g$ of the fluid column:

$$(P_2 - P_1)A = mg = \rho A h g \implies P_2 - P_1 = \rho g h$$

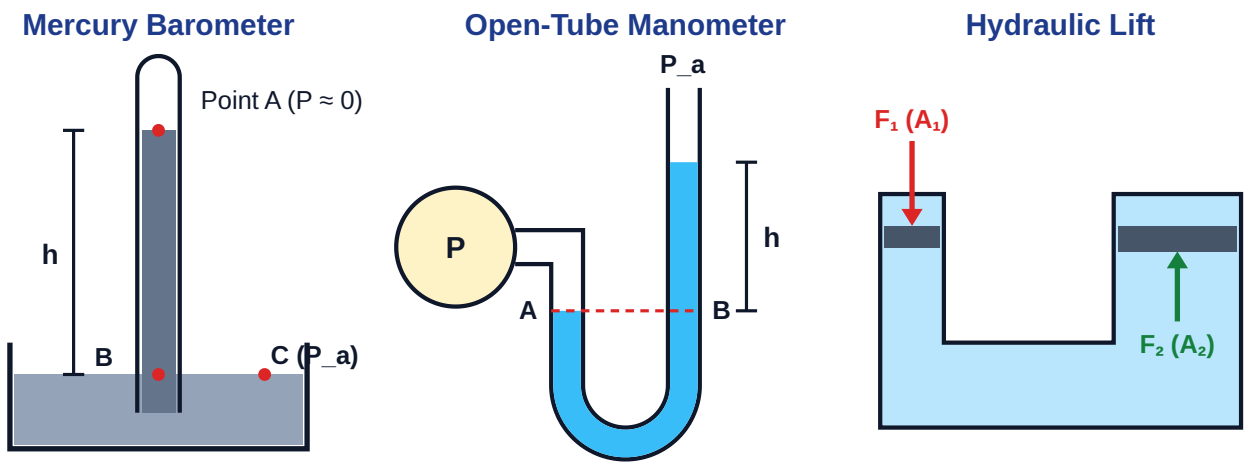
If point 1 is at the free surface of a liquid open to the atmosphere ($P_1 = P_a$, atmospheric pressure) and point 2 is at a depth h ($P_2 = P$), the **absolute pressure** P and **gauge pressure** P_g are:

$$P = P_a + \rho g h \qquad \text{and} \qquad P_g = P - P_a = \rho g h$$

Hydrostatic Paradox: Notice that the cross-sectional area A does not appear in $P = P_a + \rho g h$. Thus, liquid pressure depends only on the vertical height h of the fluid column, and *not* on the cross-sectional area, base area, or shape of the container. If three vessels A, B, and C of completely different shapes (e.g., funnel-shaped, conical, and narrow cylindrical) are connected at the bottom by a horizontal pipe and filled with water, the liquid rises to the exact same height in all three vessels despite holding very different volumes of water.

2.4 Mercury Barometer and Open-Tube Manometer

- **Mercury Barometer (Evangelista Torricelli):** A long glass tube closed at one end is filled with mercury and inverted into a trough of mercury. The space above the mercury column contains only mercury vapour, whose pressure is negligibly small ($P \approx 0$ at the top of the column). The pressure inside the tube at the level of the trough equals the atmospheric pressure P_a acting on the exposed mercury surface:
  $$P_a = \rho_{\text{Hg}} g h$$
  At sea level, $h \approx 76\,\text{cm} = 760\,\text{mm}$ of Hg, which corresponds to $1\,\text{atm}$.
- **Open-Tube Manometer:** A U-tube partially filled with a suitable liquid used to measure pressure differences. One end is open to the atmosphere (P_a) and the other is connected to a system at pressure P. Low-density liquids (such as oil) are chosen to measure small pressure differences (giving a larger, easily readable height difference h), whereas high-density liquids (such as mercury) are used for large pressure differences. The gauge pressure is $P - P_a = \rho g h$.



Tutorial Diagram 2: (Left) Torricelli's mercury barometer. (Center) Open-tube manometer measuring gauge pressure. (Right) Hydraulic lift multiplying force via Pascal's law.

2.5 Pascal's Law for Transmission of Fluid Pressure & Hydraulic Machines

Second Form of Pascal's Law: Whenever external pressure is applied on any part of a fluid contained in a vessel, it is transmitted undiminished and equally in all directions throughout the fluid and to the walls of the container.

Hydraulic Lift & Hydraulic Brakes: In a hydraulic lift, a small piston of cross-sectional area A_1 exerts a force F_1 directly on an enclosed incompressible liquid. The pressure change $P = \frac{F_1}{A_1}$ is transmitted undiminished to a larger cylinder fitted with a piston of area A_2, producing an upward force F_2:

$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2} \implies F_2 = \left(\frac{A_2}{A_1}\right) F_1$$

- **Mechanical Advantage:** The applied force is multiplied by the factor $\frac{A_2}{A_1}$.
- **Volume Conservation (Incompressibility):** If the smaller piston is pushed inward by a distance L_1, the larger piston moves outward by a distance L_2 such that the volume displaced is equal:
  $$A_1 L_1 = A_2 L_2 \implies \frac{L_2}{L_1} = \frac{A_1}{A_2}$$
  Notice that work done is conserved ($F_1 L_1 = F_2 L_2$).
- **Hydraulic Brakes:** Pressing the brake pedal moves the master piston inside the master cylinder. The resulting pressure is transmitted equally through the brake oil to pistons of larger area at all four wheels, ensuring equal braking effort on every wheel.

3. Fluid Dynamics: Streamline Flow & Bernoulli's Principle

3.1 Steady (Streamline) Flow vs. Turbulent Flow

The flow of a fluid is said to be **steady** (or **streamline**) if at any given point in space, the velocity of each passing fluid particle remains constant in time. Every particle passing a particular point behaves exactly like the previous particle that passed that point.

- **Streamline:** The smooth path taken by a fluid particle under steady flow. The tangent at any point on the curve gives the direction of the fluid velocity at that point.
- **Why two streamlines never intersect:** If two streamlines crossed each other, an oncoming fluid particle at the point of intersection could go either one way or the other (it would have two different velocity directions simultaneously), which contradicts the definition of steady flow.
- **Critical Speed & Turbulence:** Steady flow is achieved at low flow speeds. Beyond a limiting speed called the **critical speed**, the flow loses steadiness and becomes **turbulent** (characterized by irregular, fluctuating eddies and whirlpool-like regions such as white-water rapids).

3.2 Equation of Continuity

Consider steady flow of a fluid through a pipe of varying cross-section. Let the cross-sectional areas at three planes P, R, and Q bounded by the same set of streamlines be A_P, A_R, and A_Q, with corresponding fluid speeds v_P, v_R, and v_Q and densities ρ_P, ρ_R, and ρ_Q. In a small time interval Δt, conservation of mass requires that the mass of fluid entering equals the mass leaving:

$$\rho_P A_P v_P \Delta t = \rho_R A_R v_R \Delta t = \rho_Q A_Q v_Q \Delta t$$

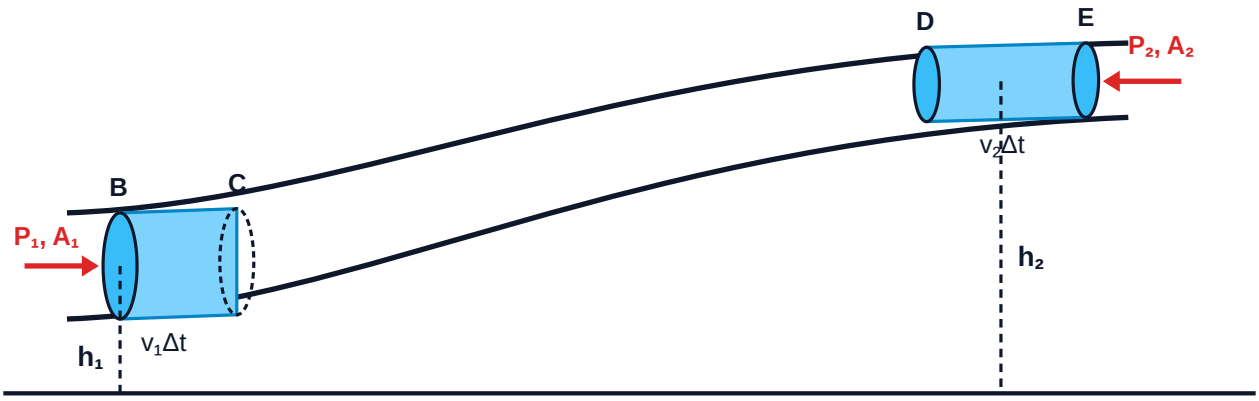
For an **incompressible fluid** ($\rho_P = \rho_R = \rho_Q$), this simplifies to the **Equation of Continuity**:

$$A_P v_P = A_R v_R = A_Q v_Q \implies Av = \text{constant}$$

Here, $Q = Av$ represents the **volume flux** or **volume flow rate** ($\mathrm{m^3\,s^{-1}}$). At narrower constrictions where streamlines are closely spaced, the flow speed v increases, and at wider portions, the flow speed decreases.

3.3 Bernoulli's Principle

Consider an ideal (incompressible and non-viscous) fluid of density ρ flowing steadily through a pipe of varying cross-section and varying height. Between region 1 (area A_1, pressure P_1, speed v_1, height h_1) and region 2 (area A_2, pressure P_2, speed v_2, height h_2), a volume $\Delta V = A_1 v_1 \Delta t = A_2 v_2 \Delta t$ of mass $\Delta m = \rho \Delta V$ moves in time Δt.



Tutorial Diagram 3: Work-energy analysis for a steady, incompressible, non-viscous fluid flowing through a pipe of varying cross-section and elevation.

By the work-energy theorem, the net work done on the fluid by pressure forces equals the sum of the change in kinetic energy and the change in gravitational potential energy:

$$W_{\text{net}} = (P_1 - P_2)\Delta V = \Delta K + \Delta U = \frac{1}{2}\rho \Delta V (v_2^2 - v_1^2) + \rho g \Delta V (h_2 - h_1)$$

Dividing throughout by ΔV and rearranging yields **Bernoulli's Equation**:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 \implies P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Statement: As we move along a streamline, the sum of the pressure (P), the kinetic energy per unit volume ($\frac{1}{2}\rho v^2$), and the potential energy per unit volume ($\rho g h$) remains constant.

Key Restrictions on Bernoulli's Equation:

1. **Non-viscous fluid ($\eta = 0$):** Assumes no kinetic energy is lost as heat due to internal friction (viscosity).
2. **Incompressible fluid ($\rho = \text{constant}$):** Elastic potential energy of the fluid is ignored.
3. **Steady (streamline) flow:** Does not apply to turbulent or non-steady flows (such as river rapids) where velocity and pressure fluctuate rapidly in time.

3.4 Applications of Bernoulli's Principle

A. Speed of Efflux: Torricelli's Law

Consider a tank containing a liquid of density ρ with a small hole in its side at height y_1 from the bottom, while the free surface of the liquid is at height y_2 (so the depth of the hole is $h = y_2 - y_1$). Let the air above the liquid surface be at pressure P, and let the pressure at the open hole be atmospheric pressure P_a. If the cross-sectional area of the tank A_2 is much larger than the area of the hole A_1 ($A_2 \gg A_1$), then by continuity $v_2 = \frac{A_1}{A_2} v_1 \approx 0$. Applying Bernoulli's equation between the top surface (2) and the hole (1):

$$P_a + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P + 0 + \rho g y_2 \implies v_1 = \sqrt{2gh + \frac{2(P - P_a)}{\rho}}$$

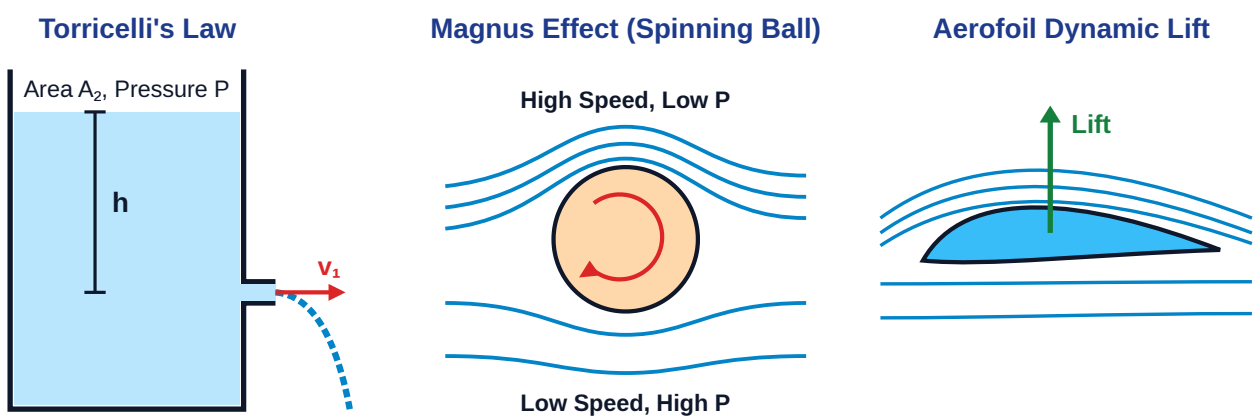
- **Closed Pressurized Tank (e.g., Rocket Propulsion):** When $P \gg P_a$ and $2gh$ can be ignored, the efflux speed is governed by the container pressure.
- **Open Tank ($P = P_a$) — Torricelli's Law:** When the tank is open to the atmosphere, $P = P_a$, and the speed of efflux is identical to the speed of a body falling freely through height h:

$$v_1 = \sqrt{2gh}$$

B. Dynamic Lift: Magnus Effect and Aerofoil

Dynamic lift is the force that acts on a body (such as an airplane wing, hydrofoil, or spinning ball) by virtue of its motion through a fluid.

- **Non-spinning ball:** Streamlines are symmetric above and below the ball. Air speeds are equal ($v_{\text{top}} = v_{\text{bottom}}$), resulting in zero pressure difference and no vertical lift.
- **Spinning ball (Magnus Effect):** A spinning ball drags a layer of air along with its rough surface. If the ball moves forward (so air streams backward relative to it) and spins clockwise, the spin adds to the relative air velocity above the ball and opposes it below. Streamlines crowd together above the ball ($v_{\text{top}} > v_{\text{bottom}}$), creating a lower pressure on top ($P_{\text{top}} < P_{\text{bottom}}$) and a net upward dynamic lift. This lateral deviation due to spin is called the **Magnus effect**.
- **Aerofoil (Aircraft Wing):** An aerofoil is shaped and oriented so that streamlines crowd together above the wing more than below it. Since $v_{\text{upper}} > v_{\text{lower}}$, Bernoulli's principle gives $P_{\text{lower}} > P_{\text{upper}}$, producing an upward dynamic lift $\Delta F = (P_{\text{lower}} - P_{\text{upper}})A$.



Tutorial Diagram 4: Applications of Bernoulli's Principle — Torricelli's efflux, Magnus effect on a spinning ball, and dynamic lift on an aircraft wing.

4. Viscosity and Stokes' Law

4.1 Coefficient of Viscosity (η)

Most real fluids offer internal resistance to motion when there is relative motion between adjacent layers of the fluid. This internal friction is called **viscosity**.

Consider a liquid layer of thickness l enclosed between a fixed bottom glass plate and a top glass plate moving with constant velocity v under a tangential force F. The fluid layer in contact with any solid surface has the same velocity as that surface (zero at the bottom plate, v at the top plate). In a small time interval Δt, the top layer moves a distance $\Delta x = v\Delta t$, producing a shear strain $\frac{\Delta x}{l}$.

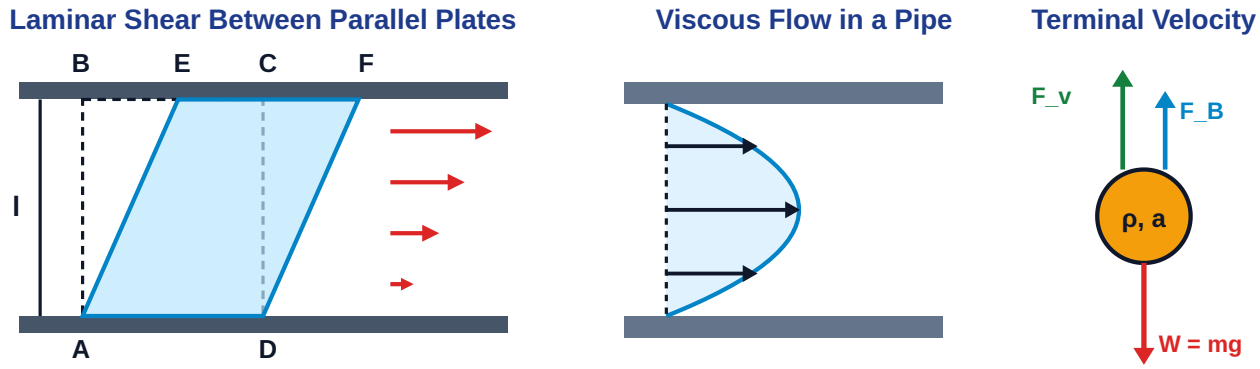
Unlike a solid (where shearing stress is proportional to shear strain), in a flowing fluid the shearing stress $\frac{F}{A}$ is proportional to the **rate of change of shear strain** (or **strain rate** / velocity gradient) $\frac{\Delta x}{l\Delta t} = \frac{v}{l}$. The **coefficient of viscosity** η is defined as:

$$\eta = \frac{\text{Shearing Stress}}{\text{Strain Rate}} = \frac{F/A}{v/l} = \frac{Fl}{vA}$$

- **SI Unit:** poiseuille (Pl), which is identical to $\mathrm{Pa\,s}$ or $\mathrm{N\,s\,m^{-2}}$ or $\mathrm{kg\,m^{-1}\,s^{-1}}$. (Note: In CGS units, $1\,\text{poise} = 0.1\,\mathrm{Pa\,s} = 100\,\mathrm{mPl}$.)
- **Dimensions:** $[\mathrm{M\,L^{-1}\,T^{-1}}]$.
- **Laminar Pipe Flow Velocity Profile:** When a viscous fluid flows steadily in a cylindrical pipe or tube, the velocity of the liquid layer is **maximum along the central axis** and decreases gradually towards the walls, where it becomes **zero**. The velocity on any coaxial cylindrical surface inside the tube is constant.
- **Temperature Dependence:**
  - In **liquids**, as temperature rises, molecules become more mobile and cohesive resistance weakens, so **viscosity decreases with increasing temperature**.
  - In **gases**, as temperature rises, the random thermal motion of atoms/molecules increases (increasing momentum transfer between layers), so **viscosity increases with increasing temperature**.
- **Physiological Note:** Blood is thicker (more viscous) than water, and the relative viscosity ($\eta_{\text{blood}}/\eta_{\text{water}}$) of blood remains constant between $0\,^\circ\mathrm{C}$ and $37\,^\circ\mathrm{C}$.

Viscosities of Common Fluids

Fluid	Temperature ($^\circ$C)	Viscosity (mPl or $10^{-3}\,\mathrm{Pa\,s}$)
Water	20	1.0
Water	100	0.3
Blood	37	2.7
Machine Oil	16	113
Machine Oil	38	34
Glycerine	20	830
Honey	–	200
Air	0	0.017
Air	40	0.019



Tutorial Diagram 5: (Left) Shear deformation of a liquid layer. (Center) Parabolic velocity distribution in a pipe. (Right) Forces on a sphere falling at terminal velocity in a viscous medium.

4.2 Stokes' Law and Terminal Velocity

When a spherical body of radius a moves with velocity v through a viscous fluid of coefficient of viscosity η, it drags the adjacent fluid layer with it and experiences a retarding viscous drag force F opposite to its motion, given by **Stokes' Law**:

$$F = 6\pi\eta a v$$

Terminal Velocity (v_t): Consider a sphere of radius a and mass density ρ falling vertically under gravity through a fluid of mass density σ and viscosity η. Initially, the sphere accelerates due to gravity. As its velocity increases, the upward retarding viscous force $6\pi\eta a v$ also increases until the sum of the viscous force and the upward buoyant force $F_B = \frac{4}{3}\pi a^3 \sigma g$ balances the downward gravitational weight $W = \frac{4}{3}\pi a^3 \rho g$. At this point, net force and acceleration become zero, and the sphere descends with a constant **terminal velocity** v_t:

$$6\pi\eta a v_t = \frac{4}{3}\pi a^3(\rho - \sigma)g \implies v_t = \frac{2a^2(\rho - \sigma)g}{9\eta}$$

Key Exam Insights on Terminal Velocity:

- $v_t \propto a^2$: Terminal velocity is directly proportional to the square of the radius of the sphere.
- $v_t \propto \frac{1}{\eta}$: Terminal velocity is inversely proportional to the viscosity of the medium.
- If $\rho < \sigma$ (such as an air bubble in water), $(\rho - \sigma)$ is negative, meaning the terminal velocity is directed **upward** (the bubble rises).

5. Surface Tension, Drops, Bubbles & Capillarity

5.1 Molecular Origin of Surface Energy and Surface Tension

A molecule well inside the bulk of a liquid is attracted equally in all directions by surrounding molecules, resulting in a large negative potential energy. However, a molecule near the free surface is surrounded by liquid molecules only in its lower half. Consequently, its negative potential energy is roughly half that of a bulk molecule. Therefore, **molecules on a liquid surface possess extra potential energy compared to molecules in the interior** (roughly half the heat of evaporation).

Because a system tends toward minimum potential energy, a liquid always tends to minimize its free surface area.

Consider a horizontal liquid film formed on a U-shaped frame with a movable slider bar of length l. When the bar is pulled outward by a small distance d by an applied force F, the work done is $W = Fd$. Because a thin film has **two free liquid-air surfaces** (top and bottom), the increase in surface area is $\Delta A = 2ld$. If S is the surface energy per unit area, conservation of energy gives:

$$S(2ld) = Fd \implies S = \frac{F}{2l} = \frac{\text{Extra Surface Energy}}{\text{Extra Surface Area}}$$

- **Definition: Surface tension (S)** is the force per unit length acting in the plane of the interface between a liquid and any other substance (or equivalently, the surface energy per unit area of the interface).
- **SI Unit & Dimensions:** $\mathrm{N\,m^{-1}}$ or $\mathrm{J\,m^{-2}}$; dimensions are $[\mathrm{M\,T^{-2}}]$.
- **Temperature Dependence:** The surface tension of a liquid generally **decreases with increasing temperature**.
- **Experimental Measurement using a Balance:** If a flat vertical glass plate of horizontal edge length l just touches a liquid surface, surface tension pulls both sides of the plate edge downward with force $2Sl$. If an extra weight $W = mg$ is required on the balance pan just to detach the plate from the liquid, then:

$$S_{\mathrm{la}} = \frac{W}{2l} = \frac{mg}{2l}$$

Surface Tension and Heats of Vaporisation of Selected Liquids

Liquid	Temperature ($^\circ$C)	Surface Tension S ($\mathrm{N\,m^{-1}}$)	Heat of Vaporisation ($\mathrm{kJ\,mol^{-1}}$)
Helium	-270	0.000239	0.115
Oxygen	-183	0.0132	7.1
Ethanol	20	0.0227	40.6
Water	20	0.0727	44.16
Mercury	20	0.4355	63.2

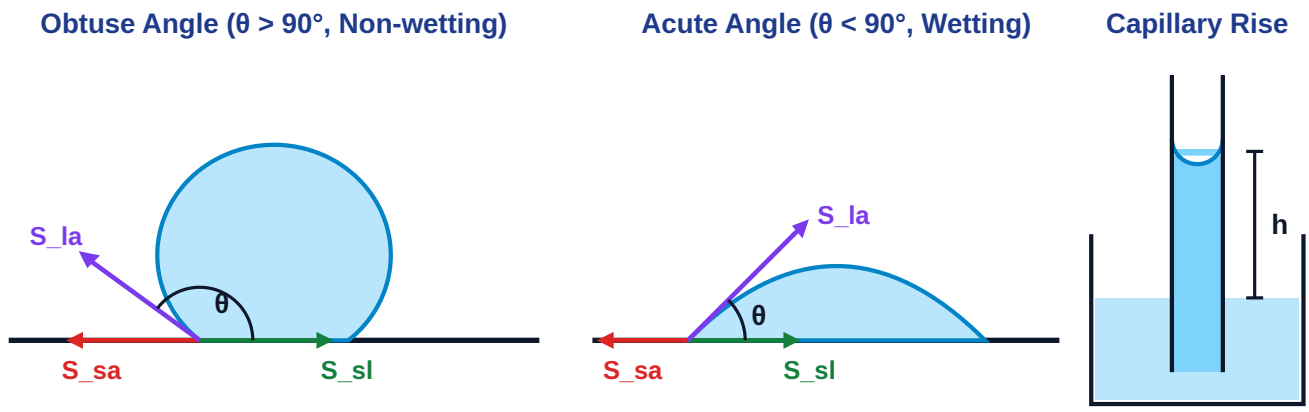
5.2 Angle of Contact (θ) and Wetting

The **angle of contact (θ)** is defined as the angle between the tangent to the liquid surface at the point of contact and the solid surface **inside the liquid**.

Let S_{la}, S_{sa}, and S_{sl} denote the interfacial tensions at the liquid-air, solid-air, and solid-liquid interfaces, respectively. For horizontal equilibrium at the line of contact of the three media:

$$S_{\text{la}} \cos\theta + S_{\text{sl}} = S_{\text{sa}} \implies \cos\theta = \frac{S_{\text{sa}} - S_{\text{sl}}}{S_{\text{la}}}$$

- **Acute Angle of Contact ($\theta < 90^\circ$, when $S_{\text{sl}} < S_{\text{sa}}$):** Liquid molecules are strongly attracted to solid molecules (adhesion > cohesion). Creating a solid-liquid interface is energetically favorable, so the liquid spreads and wets the solid (e.g., water on clean glass or plastic, kerosene on almost any surface). Soaps, detergents, and dyeing substances act as **wetting agents** by reducing θ.
- **Obtuse Angle of Contact ($\theta > 90^\circ$, when $S_{\text{sl}} > S_{\text{sa}}$):** Liquid molecules are attracted strongly to themselves and weakly to the solid (cohesion > adhesion). It costs a lot of energy to create a solid-liquid surface, so the liquid does not wet the solid and forms droplets (e.g., water on a lotus leaf or waxy surface, mercury on glass). Waterproofing agents are added to create a large obtuse angle of contact.



Tutorial Diagram 6: (Left) Obtuse angle of contact on a non-wetting surface. (Center) Acute angle of contact on a wetting surface. (Right) Capillary rise due to a concave meniscus.

5.3 Excess Pressure Inside Drops, Cavities, and Soap Bubbles

Free liquid drops and bubbles are spherical (when gravity is negligible) because for a given volume, a **sphere has the minimum surface area** and therefore minimum surface energy. Furthermore, for any curved liquid-gas interface, **the pressure on the concave side (inside) is always greater than the pressure on the convex side (outside)**.

Suppose a spherical liquid drop of radius r expands infinitesimally to $r + \Delta r$. The increase in surface area of its single liquid-air interface is $4\pi(r + \Delta r)^2 - 4\pi r^2 \approx 8\pi r \Delta r$. Equating the work done by the excess pressure $(P_i - P_o)4\pi r^2 \Delta r$ to the extra surface energy $(8\pi r \Delta r)S_{\text{la}}$:

- **Liquid Drop in Air (1 liquid-air interface):**
$$P_i - P_o = \frac{2S_{\text{la}}}{r}$$
- **Gas Bubble or Cavity Inside a Liquid (1 liquid-gas interface):**
$$P_i - P_o = \frac{2S_{\text{la}}}{r}$$
- **Soap Bubble in Air (2 liquid-air interfaces — inner and outer):**
$$P_i - P_o = \frac{4S_{\text{la}}}{r}$$

5.4 Capillary Rise (Ascent Formula)

Consider a vertical capillary tube of circular cross-section of inner radius a inserted into a vessel of liquid of density ρ and surface tension S. If the contact angle θ is acute, the meniscus is concave upward with a radius of curvature $r = \frac{a}{\cos\theta} = a\sec\theta$.

Because the air above the meniscus is on the concave side at atmospheric pressure P_a, the pressure in the liquid just below the meniscus is lower than P_a by $\Delta P = \frac{2S}{r} = \frac{2S\cos\theta}{a}$. To match the atmospheric pressure P_a at the flat liquid level outside the tube, the liquid rises in the capillary to a height h such that:

$$h\rho g = \frac{2S\cos\theta}{a} \implies h = \frac{2S\cos\theta}{\rho g a}$$

- For water in glass ($\theta \approx 0^\circ \implies \cos\theta \approx 1$), $h = \frac{2S}{\rho g a}$, so the liquid **rises** ($h \propto \frac{1}{a}$).
- For mercury in glass ($\theta > 90^\circ \implies \cos\theta < 0$), the meniscus is convex, h is negative, and the mercury level inside the capillary is **depressed (lower)** than outside.

6. Master Formula & Physical Quantities Reference Table

Summary of Physical Quantities in Fluid Mechanics

Physical Quantity	Symbol	Dimensions	SI Unit	Defining Formula / Remarks
Pressure	P	$[\mathrm{M\,L^{-1}\,T^{-2}}]$	$\mathrm{Pa}\ (\mathrm{N\,m^{-2}})$	$P = \frac{F_\perp}{A}$; Scalar quantity
Density	ρ	$[\mathrm{M\,L^{-3}}]$	$\mathrm{kg\,m^{-3}}$	$\rho = \frac{m}{V}$; Positive scalar
Relative Density (Specific Gravity)	–	Dimensionless $[\mathrm{M^0\,L^0\,T^0}]$	No unit	$\frac{\rho_{\text{substance}}}{\rho_{\text{water at 4 °C}}}$; Scalar
Volume Flux (Flow Rate)	Q	$[\mathrm{M^0\,L^3\,T^{-1}}]$	$\mathrm{m^3\,s^{-1}}$	$Q = Av = \text{constant}$ (Incompressible)
Coefficient of Viscosity	η	$[\mathrm{M\,L^{-1}\,T^{-1}}]$	$\mathrm{Pa\,s}$ or Pl	$\eta = \frac{Fl}{vA}$; Scalar
Surface Tension	S	$[\mathrm{M\,L^0\,T^{-2}}]$	$\mathrm{N\,m^{-1}}$ or $\mathrm{J\,m^{-2}}$	$S = \frac{F}{2l}$ (for 2-sided film); Scalar

7. Worked Examples (Step-by-Step)

Worked Example 1: Average Pressure on Human Femurs

Problem: The two thigh bones (femurs), each of cross-sectional area $10\,\mathrm{cm^2}$, support the upper part of a human body of mass $40\,\mathrm{kg}$. Estimate the average pressure sustained by the femurs. (Take $g = 10\,\mathrm{m\,s^{-2}}$.)

Solution:

Total cross-sectional area of the two femurs is:

$$A = 2 \times 10\,\mathrm{cm^2} = 20 \times 10^{-4}\,\mathrm{m^2} = 2 \times 10^{-3}\,\mathrm{m^2}$$

The normal downward force acting on the femurs is the weight of the upper body:

$$F = mg = 40\,\mathrm{kg} \times 10\,\mathrm{m\,s^{-2}} = 400\,\mathrm{N}$$

Therefore, the average pressure sustained is:

$$P_{\text{av}} = \frac{F}{A} = \frac{400\,\mathrm{N}}{2 \times 10^{-3}\,\mathrm{m^2}} = 2 \times 10^5\,\mathrm{N\,m^{-2}} = 2 \times 10^5\,\mathrm{Pa}$$

Worked Example 2: Pressure on a Submarine Window at Ocean Depth

Problem: At a depth of $1000\,\text{m}$ in an ocean, (a) what is the absolute pressure? (b) What is the gauge pressure? (c) Find the net force acting on a submarine window of area $20\,\text{cm} \times 20\,\text{cm}$ at this depth, given that the interior of the submarine is maintained at sea-level atmospheric pressure ($P_a = 1.01 \times 10^5\,\text{Pa}$). Take the density of sea water as $\rho = 1.03 \times 10^3\,\text{kg}\,\text{m}^{-3}$ and $g = 10\,\text{m}\,\text{s}^{-2}$.

Solution:

Given: $h = 1000\,\text{m}$, $\rho = 1.03 \times 10^3\,\text{kg}\,\text{m}^{-3}$, $g = 10\,\text{m}\,\text{s}^{-2}$.

(a) Absolute Pressure (P):

$$P = P_a + \rho g h = 1.01 \times 10^5\,\text{Pa} + (1.03 \times 10^3\,\text{kg}\,\text{m}^{-3})(10\,\text{m}\,\text{s}^{-2})(1000\,\text{m})$$

$$P = 1.01 \times 10^5\,\text{Pa} + 103 \times 10^5\,\text{Pa} = 104.01 \times 10^5\,\text{Pa} \approx 104\,\text{atm}$$

(b) Gauge Pressure (P_g):

$$P_g = P - P_a = \rho g h = 103 \times 10^5\,\text{Pa} \approx 103\,\text{atm}$$

(c) Force on the Submarine Window:

Since the outside pressure is $P = P_a + \rho g h$ and the inside pressure is P_a, the net inward pressure difference across the window equals the gauge pressure $P_g = 103 \times 10^5\,\text{Pa}$. The area of the window is $A = 0.20\,\text{m} \times 0.20\,\text{m} = 0.04\,\text{m}^2$.

$$F = P_g A = (103 \times 10^5\,\text{Pa})(0.04\,\text{m}^2) = 4.12 \times 10^5\,\text{N}$$

Worked Example 3: Hydraulic Syringes & Incompressible Displacement

Problem: Two water-filled syringes of different cross-sections (without needles) are connected by a tightly fitted water-filled rubber tube. The diameters of the smaller and larger pistons are $1.0\,\text{cm}$ and $3.0\,\text{cm}$, respectively. (a) Find the force exerted on the larger piston when a force of $10\,\text{N}$ is applied to the smaller piston. (b) If the smaller piston is pushed in through $6.0\,\text{cm}$, how much does the larger piston move out?

Solution:

(a) Force Multiplication: By Pascal's law, pressure is transmitted undiminished:

$$F_2 = \left(\frac{A_2}{A_1}\right) F_1 = \left(\frac{\pi r_2^2}{\pi r_1^2}\right) F_1 = \left(\frac{d_2}{d_1}\right)^2 F_1 = \left(\frac{3.0\,\text{cm}}{1.0\,\text{cm}}\right)^2 \times 10\,\text{N} = 9 \times 10\,\text{N} = 90\,\text{N}$$

(b) Piston Displacement: Since water is incompressible, the volume displaced inward by the smaller piston equals the volume displaced outward by the larger piston:

$$A_1 L_1 = A_2 L_2 \implies L_2 = \left(\frac{A_1}{A_2}\right) L_1 = \frac{1}{9} \times 6.0\,\text{cm} \approx 0.67\,\text{cm}$$

Worked Example 4: Dynamic Lift on a Boeing Aircraft

Problem: A fully loaded Boeing aircraft has a mass of $3.3 \times 10^5\,\mathrm{kg}$ and a total wing area of $500\,\mathrm{m}^2$. It is in level flight at a speed of $960\,\mathrm{km\,h^{-1}}$. (a) Estimate the pressure difference between the lower and upper surfaces of the wings. (b) Estimate the fractional increase in the speed of the air on the upper surface relative to the lower surface. (Take density of air $\rho = 1.2\,\mathrm{kg\,m^{-3}}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$.)

Solution:

(a) Pressure Difference (ΔP): In level flight, the upward dynamic lift $\Delta P \cdot A$ balances the weight mg:

$$\Delta P = \frac{mg}{A} = \frac{3.3 \times 10^5\,\mathrm{kg} \times 9.8\,\mathrm{m\,s^{-2}}}{500\,\mathrm{m}^2} = 6.468 \times 10^3\,\mathrm{N\,m^{-2}} \approx 6.5 \times 10^3\,\mathrm{Pa}$$

(b) Fractional Speed Increase: Ignoring the small thickness of the wing, Bernoulli's equation gives:

$$\Delta P = \frac{1}{2}\rho(v_2^2 - v_1^2) = \frac{1}{2}\rho(v_2 - v_1)(v_2 + v_1) = \rho v_{\mathrm{av}}(v_2 - v_1)$$

where $v_{\mathrm{av}} = \frac{v_1+v_2}{2} = 960\,\mathrm{km\,h^{-1}} = 960 \times \frac{5}{18}\,\mathrm{m\,s^{-1}} \approx 267\,\mathrm{m\,s^{-1}}$. Therefore, the fractional increase in speed is:

$$\frac{v_2 - v_1}{v_{\mathrm{av}}} = \frac{\Delta P}{\rho v_{\mathrm{av}}^2} = \frac{6.5 \times 10^3}{1.2 \times (267)^2} \approx 0.076 \approx 0.08 \quad (\text{or } 8\%)$$

Worked Example 5: Viscosity from a Sliding Block Experiment

Problem: A flat metal block of base area $0.10\,\mathrm{m}^2$ rests on a horizontal table separated by a liquid film of thickness $0.30\,\mathrm{mm}$. The block is connected via a string over an ideal massless, frictionless pulley to a hanging mass of $0.010\,\mathrm{kg}$. When released, the block moves at a constant speed of $0.085\,\mathrm{m\,s^{-1}}$. Find the coefficient of viscosity of the liquid. (Take $g = 9.8\,\mathrm{m\,s^{-2}}$.)

Solution:

Because the block moves with constant speed, the horizontal tension T in the string equals the weight of the suspended mass:

$$F = T = mg = 0.010\,\mathrm{kg} \times 9.8\,\mathrm{m\,s^{-2}} = 9.8 \times 10^{-2}\,\mathrm{N}$$

Using the definition of the coefficient of viscosity η with $l = 0.30\,\mathrm{mm} = 0.30 \times 10^{-3}\,\mathrm{m}$, $v = 0.085\,\mathrm{m\,s^{-1}}$, and $A = 0.10\,\mathrm{m}^2$:

$$\eta = \frac{Fl}{vA} = \frac{(9.8 \times 10^{-2}\,\mathrm{N})(0.30 \times 10^{-3}\,\mathrm{m})}{(0.085\,\mathrm{m\,s^{-1}})(0.10\,\mathrm{m}^2)} \approx 3.46 \times 10^{-3}\,\mathrm{Pa\,s}$$

Worked Example 6: Terminal Velocity of a Copper Sphere in Oil

Problem: The terminal velocity of a copper ball of radius $2.0\,\mathrm{mm}$ falling through a tank of oil at $20\,^\circ\mathrm{C}$ is $6.5\,\mathrm{cm\,s^{-1}}$. Compute the viscosity of the oil at $20\,^\circ\mathrm{C}$. Given: density of oil $\sigma = 1.5 \times 10^3\,\mathrm{kg\,m^{-3}}$, density of copper $\rho = 8.9 \times 10^3\,\mathrm{kg\,m^{-3}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$.

Solution:

Given: $v_t = 6.5 \times 10^{-2}\,\mathrm{m\,s^{-1}}$, $a = 2.0 \times 10^{-3}\,\mathrm{m}$, and $\rho - \sigma = (8.9 - 1.5) \times 10^3 = 7.4 \times 10^3\,\mathrm{kg\,m^{-3}}$.

From the terminal velocity formula $v_t = \frac{2a^2(\rho-\sigma)g}{9\eta}$, we solve for η:

$$\eta = \frac{2a^2(\rho - \sigma)g}{9v_t} = \frac{2 \times (2.0 \times 10^{-3}\,\mathrm{m})^2 \times (7.4 \times 10^3\,\mathrm{kg\,m^{-3}}) \times 9.8\,\mathrm{m\,s^{-2}}}{9 \times 6.5 \times 10^{-2}\,\mathrm{m\,s^{-1}}} \approx 9.9 \times 10^{-1}\,\mathrm{Pa\,s}$$

Worked Example 7: Blowing a Hemispherical Bubble at the End of a Submerged Capillary

Problem: The lower end of a capillary tube of diameter $2.00\,\text{mm}$ is dipped $8.00\,\text{cm}$ below the surface of water in a beaker. What is the pressure required in the tube to blow a hemispherical air bubble at its lower end in water? Also calculate the excess pressure. (Surface tension of water $S = 7.30 \times 10^{-2}\,\text{N m}^{-1}$, $1\,\text{atm} = 1.01 \times 10^5\,\text{Pa}$, $\rho_{\text{water}} = 1000\,\text{kg m}^{-3}$, $g = 9.80\,\text{m s}^{-2}$.)

Solution:

For a hemispherical bubble formed at the tip of the capillary tube, the radius of the bubble equals the radius of the capillary tube: $r = \frac{2.00\,\text{mm}}{2} = 1.00\,\text{mm} = 1.00 \times 10^{-3}\,\text{m}$.

Since an air bubble inside water has only **one** liquid-gas interface, the excess pressure is:

$$\Delta P = P_i - P_o = \frac{2S}{r} = \frac{2 \times 7.30 \times 10^{-2}\,\text{N m}^{-1}}{1.00 \times 10^{-3}\,\text{m}} = 146\,\text{Pa}$$

The pressure P_o outside the bubble at depth $h = 8.00\,\text{cm} = 0.08\,\text{m}$ is:

$$P_o = P_a + \rho g h = 1.01 \times 10^5\,\text{Pa} + (1000\,\text{kg m}^{-3})(9.80\,\text{m s}^{-2})(0.08\,\text{m}) = 1.01784 \times 10^5\,\text{Pa}$$

Therefore, the absolute pressure P_i required inside the tube is:

$$P_i = P_o + \frac{2S}{r} = 1.01784 \times 10^5\,\text{Pa} + 146\,\text{Pa} = 1.0193 \times 10^5\,\text{Pa} \approx 1.02 \times 10^5\,\text{Pa}$$

8. NEET UG Exam-Style Practice Questions

The following original practice questions are strictly aligned with the NEET UG Physics examination format (Single-Correct Four-Option Multiple Choice Questions, including numerical, conceptual, statement-based, assertion-reason, and matching types). Complete worked solutions follow each question.

Practice Question 1 (Hydrostatics & U-Tube Equilibrium)

A vertical U-tube contains water and methylated spirit separated by mercury. Initially, the mercury columns in the two arms are at the exact same horizontal level when there is a $10.0\,\text{cm}$ column of water in one arm and a $12.5\,\text{cm}$ column of spirit in the other. What is the specific gravity (relative density) of the methylated spirit?

(1) 0.75

(2) 0.80

(3) 0.85

(4) 1.25

Correct Answer: (2)

Worked Solution: Since the mercury levels in both arms are at the same horizontal plane, the pressures at the mercury-liquid interfaces in both arms must be equal:

$$P_a + \rho_w g h_w = P_a + \rho_s g h_s \implies \rho_w h_w = \rho_s h_s$$

Therefore, the specific gravity (relative density) of spirit is:

$$\frac{\rho_s}{\rho_w} = \frac{h_w}{h_s} = \frac{10.0\,\text{cm}}{12.5\,\text{cm}} = 0.80$$

Practice Question 2 (Differential Mercury Level in U-Tube)

In the U-tube described in Practice Question 1 (initially having $10.0\,\text{cm}$ of water of specific gravity 1.0 in one arm and $12.5\,\text{cm}$ of spirit of specific gravity 0.80 in the other, with mercury levels equal), an additional $15.0\,\text{cm}$ of water and $15.0\,\text{cm}$ of spirit are poured into their respective arms. If the specific gravity of mercury is 13.6, what is the new difference in the levels of mercury between the two arms?

(1) $0.221\,\text{cm}$

(2) $0.441\,\text{cm}$

(3) $0.110\,\text{cm}$

(4) $0.300\,\text{cm}$

Correct Answer: (1)

Worked Solution: Total height of water column is $h'_w = 10.0 + 15.0 = 25.0\,\text{cm}$, and total height of spirit column is $h'_s = 12.5 + 15.0 = 27.5\,\text{cm}$. Since the initial $10.0\,\text{cm}$ of water balanced $12.5\,\text{cm}$ of spirit, the extra pressure difference is due to the added $15.0\,\text{cm}$ columns. Equating pressures at the lower mercury interface (let h_{Hg} be the difference in mercury levels):

$$\rho_{\text{Hg}} g h_{\text{Hg}} = \rho_w g h'_w - \rho_s g h'_s$$

Dividing by $\rho_w g$:

$$13.6\, h_{\text{Hg}} = (1.0)(25.0\,\text{cm}) - (0.80)(27.5\,\text{cm}) = 25.0 - 22.0 = 3.0\,\text{cm}$$

$$h_{\text{Hg}} = \frac{3.0}{13.6}\,\text{cm} \approx 0.221\,\text{cm}$$

Practice Question 3 (Contact Force and Pressure)

A $50\,\text{kg}$ girl wearing high-heel shoes balances her entire weight on a single circular heel of diameter $1.0\,\text{cm}$. What is the pressure exerted by the heel on the horizontal floor? (Take $g = 9.8\,\text{m}\,\text{s}^{-2}$.)

(1) $6.24 \times 10^6\,\text{Pa}$

(2) $1.56 \times 10^6\,\text{Pa}$

(3) $3.12 \times 10^6\,\text{Pa}$

(4) $4.90 \times 10^5\,\text{Pa}$

Correct Answer: (1)

Worked Solution: Radius of the circular heel is $r = \frac{d}{2} = 0.5\,\text{cm} = 5.0 \times 10^{-3}\,\text{m}$. Area of the heel is:

$$A = \pi r^2 = 3.1416 \times (5.0 \times 10^{-3}\,\text{m})^2 \approx 7.854 \times 10^{-5}\,\text{m}^2$$

Normal force is $F = mg = 50 \times 9.8 = 490\,\text{N}$. Thus, pressure is:

$$P = \frac{F}{A} = \frac{490\,\text{N}}{7.854 \times 10^{-5}\,\text{m}^2} \approx 6.24 \times 10^6\,\text{Pa}$$

Practice Question 4 (Barometric Height of French Wine)

Torricelli's barometer used mercury ($\rho_{\text{Hg}} = 13.6 \times 10^3\,\text{kg m}^{-3}$). Pascal duplicated it using French wine of density $984\,\text{kg m}^{-3}$. What is the height of the wine column required to balance normal sea-level atmospheric pressure ($1.013 \times 10^5\,\text{Pa}$)? (Take $g = 9.8\,\text{m s}^{-2}$.)

(1) $8.4\,\text{m}$

(2) $10.5\,\text{m}$

(3) $13.6\,\text{m}$

(4) $9.84\,\text{m}$

Correct Answer: (2)

Worked Solution: Using the barometer relation $P_a = \rho_{\text{wine}} g h$:

$$h = \frac{P_a}{\rho_{\text{wine}} g} = \frac{1.013 \times 10^5\,\text{Pa}}{984\,\text{kg m}^{-3} \times 9.8\,\text{m s}^{-2}} = \frac{1.013 \times 10^5}{9643.2}\,\text{m} \approx 10.5\,\text{m}$$

Practice Question 5 (Hydraulic Automobile Lift)

A hydraulic automobile lift is designed to lift cars with a maximum mass of $3000\,\text{kg}$. The area of cross-section of the piston carrying the load is $425\,\text{cm}^2$. What maximum pressure would the smaller piston have to bear? (Take $g = 9.8\,\text{m s}^{-2}$.)

(1) $6.92 \times 10^5\,\text{Pa}$

(2) $3.46 \times 10^5\,\text{Pa}$

(3) $1.38 \times 10^6\,\text{Pa}$

(4) $7.06 \times 10^4\,\text{Pa}$

Correct Answer: (1)

Worked Solution: By Pascal's law, the pressure in the fluid at the smaller piston (at the same level) equals the pressure exerted on the larger load-carrying piston:

$$P = \frac{F_{\max}}{A_2} = \frac{m_{\max} g}{A_2} = \frac{3000\,\text{kg} \times 9.8\,\text{m s}^{-2}}{425 \times 10^{-4}\,\text{m}^2} = \frac{29400}{0.0425}\,\text{Pa} \approx 6.92 \times 10^5\,\text{Pa}$$

Practice Question 6 (Equation of Continuity — Spray Pump)

The cylindrical tube of a spray pump has a cross-sectional area of $8.0\,\mathrm{cm}^2$, one end of which has 40 fine circular holes, each of diameter $1.0\,\mathrm{mm}$. If the liquid flow speed inside the tube is $1.5\,\mathrm{m\,min}^{-1}$, what is the speed of ejection of the liquid through the holes?

(1) $0.64\,\mathrm{m\,s}^{-1}$

(2) $1.27\,\mathrm{m\,s}^{-1}$

(3) $2.54\,\mathrm{m\,s}^{-1}$

(4) $0.32\,\mathrm{m\,s}^{-1}$

Correct Answer: (1)

Worked Solution: Area of the tube is $A_1 = 8.0\,\mathrm{cm}^2 = 8.0 \times 10^{-4}\,\mathrm{m}^2$. Speed of liquid inside the tube is:

$$v_1 = \frac{1.5\,\mathrm{m}}{60\,\mathrm{s}} = 0.025\,\mathrm{m\,s}^{-1}$$

Radius of each hole is $r = 0.5\,\mathrm{mm} = 0.5 \times 10^{-3}\,\mathrm{m}$. Total area of all 40 holes is:

$$A_2 = 40 \times \pi r^2 = 40 \times 3.1416 \times (0.5 \times 10^{-3}\,\mathrm{m})^2 = 3.1416 \times 10^{-5}\,\mathrm{m}^2$$

By the Equation of Continuity ($A_1 v_1 = A_2 v_2$):

$$v_2 = \frac{A_1 v_1}{A_2} = \frac{(8.0 \times 10^{-4}\,\mathrm{m}^2)(0.025\,\mathrm{m\,s}^{-1})}{3.1416 \times 10^{-5}\,\mathrm{m}^2} = \frac{2.0 \times 10^{-5}}{3.1416 \times 10^{-5}}\,\mathrm{m\,s}^{-1} \approx 0.637\,\mathrm{m\,s}^{-1} \approx 0.64\,\mathrm{m\,s}^{-1}$$

Practice Question 7 (Wind Tunnel Aerofoil Lift)

In a test experiment on a model aeroplane in a wind tunnel, the air flow speeds on the upper and lower surfaces of the wing are $70\,\mathrm{m\,s}^{-1}$ and $63\,\mathrm{m\,s}^{-1}$, respectively. What is the dynamic lift on the wing if its area is $2.5\,\mathrm{m}^2$? (Take the density of air to be $1.3\,\mathrm{kg\,m}^{-3}$.)

(1) $1.51 \times 10^3\,\mathrm{N}$

(2) $3.02 \times 10^3\,\mathrm{N}$

(3) $6.05 \times 10^2\,\mathrm{N}$

(4) $7.56 \times 10^2\,\mathrm{N}$

Correct Answer: (1)

Worked Solution: Using Bernoulli's principle, the pressure difference between the lower and upper surfaces of the wing is:

$$\Delta P = P_{\text{lower}} - P_{\text{upper}} = \frac{1}{2}\rho(v_{\text{upper}}^2 - v_{\text{lower}}^2) = \frac{1}{2}(1.3\,\mathrm{kg\,m}^{-3})(70^2 - 63^2)\,\mathrm{m}^2\,\mathrm{s}^{-2}$$

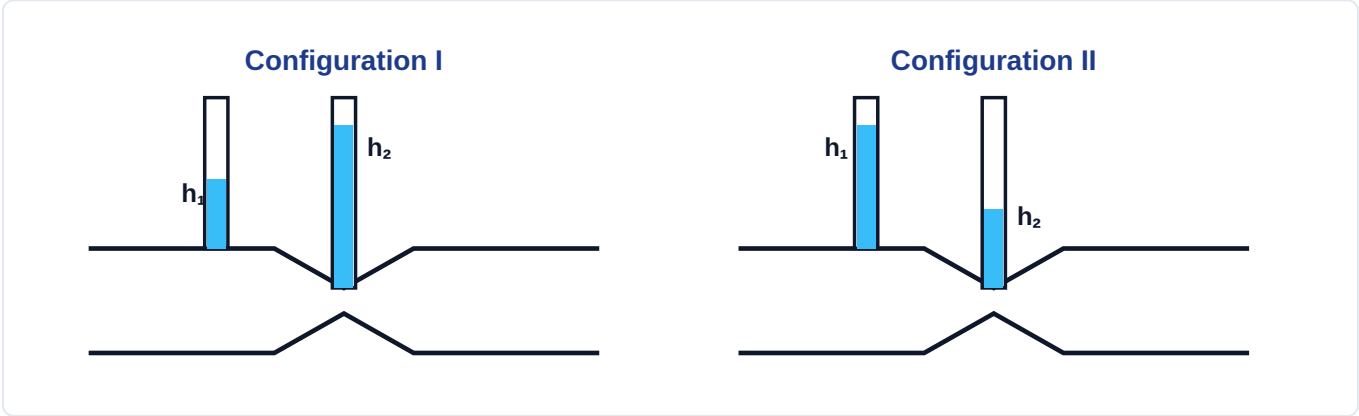
$$\Delta P = 0.65 \times (4900 - 3969) = 0.65 \times 931 = 605.15\,\mathrm{Pa}$$

Therefore, the upward lift force on the wing of area $A = 2.5\,\mathrm{m}^2$ is:

$$F_{\text{lift}} = \Delta P \times A = 605.15\,\mathrm{Pa} \times 2.5\,\mathrm{m}^2 \approx 1512.9\,\mathrm{N} \approx 1.51 \times 10^3\,\mathrm{N}$$

Practice Question 8 (Constriction Pressure & Piezometer Tubes)

A non-viscous, incompressible liquid flows steadily through a horizontal pipe that has a wide section of cross-sectional area A_1 followed by a narrow constriction (throat) of cross-sectional area A_2 ($A_2 < A_1$). Vertical open glass tubes (piezometers) are attached at the wide section and at the narrow throat to indicate static pressure via liquid column heights h_1 and h_2, respectively. Which of the following correctly describes the heights of the liquid columns?



(1) Configuration I is correct because the streamlines crowd at the neck, increasing pressure ($h_2 > h_1$).

(2) Configuration II is correct because by continuity $v_2 > v_1$ at the narrow neck, and by Bernoulli's principle the pressure is lower at the neck ($h_2 < h_1$).

(3) Both Configurations I and II are incorrect because Pascal's law requires $h_1 = h_2$ in any horizontal pipe.

(4) Configuration I is correct for liquids while Configuration II is correct for gases.

Correct Answer: (2)

Worked Solution: By the Equation of Continuity ($A_1 v_1 = A_2 v_2$), since $A_2 < A_1$ at the narrow constriction, the flow speed is higher at the constriction ($v_2 > v_1$). By Bernoulli's equation for a horizontal pipe ($P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$), a higher speed corresponds to a lower pressure ($P_2 < P_1$). Therefore, the liquid column in the vertical tube at the narrow neck must be lower than at the wider section ($h_2 < h_1$), making Configuration II correct and Configuration I incorrect.

Practice Question 9 (Surface Tension of a Soap Film)

A U-shaped wire is dipped in a soap solution and removed. The thin soap film formed between the wire and a light horizontal slider of length $30\,\text{cm}$ supports a total weight of $1.5 \times 10^{-2}\,\text{N}$ (which includes the small weight of the slider). What is the surface tension of the soap film?

(1) $5.0 \times 10^{-2}\,\text{N m}^{-1}$

(2) $2.5 \times 10^{-2}\,\text{N m}^{-1}$

(3) $1.25 \times 10^{-2}\,\text{N m}^{-1}$

(4) $4.5 \times 10^{-3}\,\text{N m}^{-1}$

Correct Answer: (2)

Worked Solution: A soap film has **two** free liquid-air surfaces. Therefore, the upward surface tension force acting on the slider of length $l = 30\,\text{cm} = 0.30\,\text{m}$ is $F = 2Sl$. At equilibrium:

$$2Sl = W \implies S = \frac{W}{2l} = \frac{1.5 \times 10^{-2}\,\text{N}}{2 \times 0.30\,\text{m}} = 2.5 \times 10^{-2}\,\text{N m}^{-1}$$

Practice Question 10 (Surface Tension Independence of Film Area & Shape)

Three wire frames (a), (b), and (c) hold thin films of the exact same liquid at the same temperature, and each frame has a horizontal movable bottom slider of the exact same length $l = 40\,\text{cm}$. Frame (a) has a short rectangular film supporting a weight $W_a = 4.5 \times 10^{-2}\,\text{N}$ in equilibrium. Frame (b) has a tall rectangular film of twice the surface area, and Frame (c) has a pentagon-shaped top with the same $40\,\text{cm}$ bottom slider. What weights W_b and W_c are supported in equilibrium by the films in (b) and (c)?

(1) $W_b = 9.0 \times 10^{-2}\,\text{N}$ and $W_c = 4.5 \times 10^{-2}\,\text{N}$

(2) $W_b = 4.5 \times 10^{-2}\,\text{N}$ and $W_c = 4.5 \times 10^{-2}\,\text{N}$

(3) $W_b = 2.25 \times 10^{-2}\,\text{N}$ and $W_c = 3.0 \times 10^{-2}\,\text{N}$

(4) $W_b = 9.0 \times 10^{-2}\,\text{N}$ and $W_c = 6.0 \times 10^{-2}\,\text{N}$

Correct Answer: (2)

Worked Solution: The surface tension S of a liquid depends only on the nature of the liquid and the temperature, and is completely **independent of the area or shape of the film**. Because the length of the bottom slider $l = 40\,\text{cm}$ is the same in all three cases, the upward force $F = 2Sl$ is identical in (a), (b), and (c). Hence, $W_b = W_c = 4.5 \times 10^{-2}\,\text{N}$.

Practice Question 11 (Pressure Inside a Mercury Drop)

What are the excess pressure and the total absolute pressure inside a spherical drop of mercury of radius $3.00\,\text{mm}$ at room temperature ($20\,^\circ\text{C}$)? Given: surface tension of mercury at $20\,^\circ\text{C}$ is $S = 4.65 \times 10^{-1}\,\text{N m}^{-1}$ and atmospheric pressure is $P_a = 1.01 \times 10^5\,\text{Pa}$.

(1) Excess pressure $= 310\,\text{Pa}$; Total pressure $\approx 1.013 \times 10^5\,\text{Pa}$

(2) Excess pressure $= 620\,\text{Pa}$; Total pressure $\approx 1.016 \times 10^5\,\text{Pa}$

(3) Excess pressure $= 155\,\text{Pa}$; Total pressure $\approx 1.012 \times 10^5\,\text{Pa}$

(4) Excess pressure $= 930\,\text{Pa}$; Total pressure $\approx 1.019 \times 10^5\,\text{Pa}$

Correct Answer: (1)

Worked Solution: A liquid drop has a single liquid-air interface. With $r = 3.00 \times 10^{-3}\,\text{m}$ and $S = 0.465\,\text{N m}^{-1}$:

$$\Delta P = P_i - P_o = \frac{2S}{r} = \frac{2 \times 4.65 \times 10^{-1}\,\text{N m}^{-1}}{3.00 \times 10^{-3}\,\text{m}} = 310\,\text{Pa}$$

Total internal pressure is:

$$P_i = P_a + \Delta P = 1.01 \times 10^5\,\text{Pa} + 310\,\text{Pa} = 1.0131 \times 10^5\,\text{Pa} \approx 1.013 \times 10^5\,\text{Pa}$$

Practice Question 12 (Soap Bubble vs. Submerged Air Bubble)

The surface tension of a soap solution at $20\,^{\circ}\mathrm{C}$ is $2.50 \times 10^{-2}\,\mathrm{N\,m^{-1}}$ and its relative density is 1.20. First, a soap bubble of radius $5.00\,\mathrm{mm}$ is blown in air. Second, an air bubble of the same radius ($5.00\,\mathrm{mm}$) is formed at a depth of $40.0\,\mathrm{cm}$ inside a container filled with the soap solution. What are (i) the excess pressure inside the soap bubble in air, and (ii) the total absolute pressure inside the submerged air bubble? (Take $1\,\mathrm{atm} = 1.01 \times 10^5\,\mathrm{Pa}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$.)

(1) (i) $10.0\,\mathrm{Pa}$, (ii) $1.057 \times 10^5\,\mathrm{Pa}$

(2) (i) $20.0\,\mathrm{Pa}$, (ii) $1.057 \times 10^5\,\mathrm{Pa}$

(3) (i) $20.0\,\mathrm{Pa}$, (ii) $1.010 \times 10^5\,\mathrm{Pa}$

(4) (i) $40.0\,\mathrm{Pa}$, (ii) $1.065 \times 10^5\,\mathrm{Pa}$

Correct Answer: (2)

Worked Solution:

(i) Soap bubble in air (2 interfaces): With $r = 5.00 \times 10^{-3}\,\mathrm{m}$ and $S = 2.50 \times 10^{-2}\,\mathrm{N\,m^{-1}}$:

$$\Delta P_{\text{soap}} = \frac{4S}{r} = \frac{4 \times 2.50 \times 10^{-2}\,\mathrm{N\,m^{-1}}}{5.00 \times 10^{-3}\,\mathrm{m}} = 20.0\,\mathrm{Pa}$$

(ii) Air bubble submerged at $h = 0.40\,\mathrm{m}$ inside soap solution (1 interface):

Density of soap solution is $\rho = 1.20 \times 10^3\,\mathrm{kg\,m^{-3}}$. Excess pressure across its single interface is $\Delta P_{\text{air}} = \frac{2S}{r} = 10.0\,\mathrm{Pa}$. Total pressure inside the submerged air bubble is:

$$P_i = P_a + \rho g h + \frac{2S}{r} = 1.01 \times 10^5 + (1.20 \times 10^3)(9.8)(0.40) + 10.0$$

$$P_i = 1.01 \times 10^5 + 4704 + 10.0 = 1.05714 \times 10^5\,\mathrm{Pa} \approx 1.057 \times 10^5\,\mathrm{Pa}$$

Practice Question 13 (Assertion–Reason: Temperature Dependence of Viscosity & Surface Tension)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): When the temperature of water is raised from $20\ ^\circ\mathrm{C}$ to $80\ ^\circ\mathrm{C}$, both its coefficient of viscosity and its surface tension decrease, whereas when the temperature of air is raised from $0\ ^\circ\mathrm{C}$ to $40\ ^\circ\mathrm{C}$, its coefficient of viscosity increases.

Reason (R): A rise in temperature increases the mobility of molecules in a liquid (reducing cohesive internal resistance and surface energy), whereas in a gas it increases the random thermal motion of molecules.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution: In liquids, increasing temperature increases molecular mobility, causing both viscosity (η) and surface tension (S) to fall (for water, η drops from $1.0\ \mathrm{mPl}$ at $20\ ^\circ\mathrm{C}$ to $0.3\ \mathrm{mPl}$ at $100\ ^\circ\mathrm{C}$). In gases, increasing temperature enhances random thermal motion of atoms/molecules, increasing momentum exchange between layers and causing η to increase (for air, η rises from $0.017\ \mathrm{mPl}$ at $0\ ^\circ\mathrm{C}$ to $0.019\ \mathrm{mPl}$ at $40\ ^\circ\mathrm{C}$). Thus both (A) and (R) are true and (R) correctly explains (A).

Practice Question 14 (Statement-Based Question: Conceptual Fluid Mechanics)

Consider the following two statements regarding fluid mechanics:

Statement I: While administering an injection, the size (radius) of the needle of a syringe controls the volume flow rate better than the thumb pressure exerted by the doctor, and to keep a piece of paper horizontal one should blow over it rather than under it.

Statement II: It makes a significant difference if one uses gauge pressures instead of absolute pressures on both sides when applying Bernoulli's equation between two points where the atmospheric pressure is the same.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

Correct Answer: (3)

Worked Solution:

- **Statement I is correct:** By the equation of continuity ($Av = \text{constant}$) and Bernoulli's principle, flow speed and flow rate are highly sensitive to the cross-sectional area of the needle constriction, whereas thumb pressure enters only under a square root (or linearly in viscous flow vs fourth power of radius). Also, blowing over a paper increases air speed above it, reducing pressure on top by Bernoulli's principle so atmospheric pressure below keeps it horizontal.
- **Statement II is incorrect:** Since $P_{\text{abs}} = P_a + P_{\text{gauge}}$, if atmospheric pressure P_a is the same at both points, P_a cancels from both sides of Bernoulli's equation ($P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$). Hence, using gauge pressure instead of absolute pressure makes no difference at all unless P_a differs significantly between the two locations.

Practice Question 15 (Matching List — Physical Quantities, Laws & Dimensions)

Match **List I** (Fluid Phenomenon / Physical Quantity) with **List II** (Governing Principle / Dimensional Formula):

List I	List II
A. Increase in flow speed at a pipe constriction	I. $[\mathrm{M\,L^{-1}\,T^{-1}}]$
B. Coefficient of viscosity (η)	II. $[\mathrm{M\,L^{0}\,T^{-2}}]$
C. Surface tension (S)	III. Conservation of mass in incompressible fluid
D. Terminal velocity of a raindrop (v_t)	IV. Proportional to the square of the radius (a^2)

Choose the correct answer from the options given below:

(1) A–III, B–I, C–II, D–IV

(2) A–III, B–II, C–I, D–IV

(3) A–IV, B–I, C–II, D–III

(4) A–I, B–III, C–IV, D–II

Correct Answer: (1)

Worked Solution:

- **A** $\rightarrow$ **III:** The increase in flow speed at a constriction ($Av = \text{constant}$) follows directly from the conservation of mass for an incompressible fluid.
- **B** $\rightarrow$ **I:** Coefficient of viscosity $\eta = \frac{Fl}{vA}$ has dimensions $\frac{[\mathrm{M\,L\,T^{-2}}][\mathrm{L}]}{[\mathrm{L\,T^{-1}}][\mathrm{L^2}]} = [\mathrm{M\,L^{-1}\,T^{-1}}]$.
- **C** $\rightarrow$ **II:** Surface tension $S = \frac{F}{l}$ has dimensions $\frac{[\mathrm{M\,L\,T^{-2}}]}{[\mathrm{L}]} = [\mathrm{M\,L^{0}\,T^{-2}}]$.
- **D** $\rightarrow$ **IV:** From Stokes' law, terminal velocity $v_t = \frac{2a^2(\rho-\sigma)g}{9\eta} \propto a^2$.

Practice Question 16 (Coalescence of Drops & Terminal Velocity)

Eight identical spherical raindrops, each falling through air with a steady terminal velocity of $5\,\text{cm}\,\text{s}^{-1}$, coalesce (combine) to form a single larger spherical drop. Neglecting any change in air density or viscosity, what is the new terminal velocity of the combined drop?

(1) $10\,\text{cm}\,\text{s}^{-1}$

(2) $20\,\text{cm}\,\text{s}^{-1}$

(3) $40\,\text{cm}\,\text{s}^{-1}$

(4) $15\,\text{cm}\,\text{s}^{-1}$

Correct Answer: (2)

Worked Solution: Let a be the radius of each small drop and R be the radius of the combined drop. By conservation of volume:

$$\frac{4}{3}\pi R^3 = 8 \times \left(\frac{4}{3}\pi a^3\right) \implies R = 2a$$

Since terminal velocity is proportional to the square of the radius ($v_t \propto r^2$):

$$\frac{v_t'}{v_t} = \left(\frac{R}{a}\right)^2 = 2^2 = 4 \implies v_t' = 4 \times 5\,\text{cm}\,\text{s}^{-1} = 20\,\text{cm}\,\text{s}^{-1}$$

9. Common Exam Pitfalls & Conceptual Clarifications

Pitfall 1: Treating Pressure as a Vector Quantity
Because pressure is defined as "force divided by area", students often mistakenly treat pressure as a vector. Remember: only the scalar magnitude of the force component *normal* to the surface appears in $P = \frac{F_\perp}{A}$. Hydrostatic pressure at a point acts equally in all directions and has no unique direction assigned to it.

Pitfall 2: Confusing $2S/r$ and $4S/r$ for Bubbles
Always count the number of liquid-gas interfaces! A **liquid drop in air** and an **air bubble submerged inside a liquid** each have only **one** interface, so $\Delta P = \frac{2S}{r}$. A **soap bubble in air** is a thin spherical shell of liquid with air both inside and outside, giving **two** interfaces and $\Delta P = \frac{4S}{r}$.

Pitfall 3: Applying $\Delta P = \rho g h$ to Compressible Gases Over Large Heights
The formula $P = P_a + \rho g h$ assumes that the fluid density ρ is uniform (incompressible). This holds very well for liquids. However, gases are highly compressible and their density decreases rapidly with altitude (and g also varies). For example, although a uniform-density calculation gives an atmosphere height of only $\approx 8\,\text{km}$, Earth's actual atmosphere extends over $100\,\text{km}$, and atmospheric pressure drops to nearly half its sea-level value at an altitude of about $6\,\text{km}$.

Pitfall 4: Misidentifying the Cause of Speed Increase at a Constriction
In steady incompressible flow, the increase in speed at a pipe constriction ($Av = \text{constant}$) is a direct consequence of **conservation of mass** (Equation of Continuity), *not* Bernoulli's principle. Bernoulli's principle (conservation of energy) then explains why the pressure drops where the speed increases.

10. Concise Final Revision Checklist

- **Hydrostatics:** $P = P_a + \rho g h$; Gauge pressure $P_g = P - P_a = \rho g h$. Blood pressure in humans is greater at the feet than at the brain due to the extra height h of the blood column.
- **Hydraulic Machines:** $\frac{F_1}{A_1} = \frac{F_2}{A_2}$ and $A_1 L_1 = A_2 L_2$.
- **Continuity & Bernoulli:** $A_1 v_1 = A_2 v_2$ and $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$ (valid for steady, incompressible, non-viscous flow).
- **Torricelli's Efflux Law:** $v_1 = \sqrt{2gh + \frac{2(P - P_a)}{\rho}}$; for an open tank ($P = P_a$), $v_1 = \sqrt{2gh}$. Outflow of fluid from a small hole also imparts a backward reaction thrust on the vessel.
- **Viscosity & Stokes' Law:** $\eta = \frac{Fl}{vA}$; $F_{\text{drag}} = 6\pi\eta a v$; Terminal velocity $v_t = \frac{2a^2(\rho - \sigma)g}{9\eta}$.
- **Surface Tension & Capillarity:** $S = \frac{F}{2l}$ (for a 2-sided film); $\cos\theta = \frac{S_{\text{sa}} - S_{\text{sl}}}{S_{\text{la}}}$; Capillary rise $h = \frac{2S\cos\theta}{\rho g a}$.